SDC HACKATHON 2026 · DOMAIN 2 · ORBYTE SMART SME CHALLENGE

Orbyte SlipMatch

Instant Chat-to-Bill Generator & Automated DuitNow Receipt Validator

Team Catalyst



THE CONTEXT

Malaysia runs on micro businesses, and they get paid by transfer

96%

of all businesses in Malaysia are MSMEs, almost 1.3 million firms

77%+

of MSMEs are micro enterprises

3B

DuitNow QR transactions in 2025, double the year before

Sources: SME Corp briefing via Bernama / Malay Mail (31 Jul 2026) · Bank Negara Malaysia Annual Report 2025

THE PROBLEM

Many small sellers do it all by hand, every day

For every single order, the owner has to:

1. Read the WhatsApp order and write it down
2. Wait for the payment screenshot
3. Open the bank app to see if the money came in
4. Find the right order and mark it as paid

1 in 3

Malaysian SMEs still run their business on scattered tools, like Excel files and many WhatsApp groups

1 in 100

entries typed by hand comes out wrong, and checking by eye does not catch them

Sources: The Borneo Post (4 Jan 2026) · Barchard & Pace (2011), Computers in Human Behavior

THE BIGGER RISK

RM123.7 million

lost to e-commerce fraud in Malaysia, January to November 2025, across 14,881 cases

Fake payment proof: the buyer sends an edited screenshot and no money arrives. One Port Dickson trader lost RM780 to a fake transfer receipt in 2025.

Sources: PDRM via Bernama (8 Dec 2025) · The Vibes (17 Sep 2025)

OUR SOLUTION

SliplipMatchMatch: from WhatsApp chat to verified payment

1 · Chat-to-Bill

- Paste a WhatsApp order in Bahasa Melayu, English or Manglish.
- AI-assisted extraction identifies customer details and requested items
- Menu prices and billing rules calculate the total.
- Missing information is flagged for review.

2 · Receipt Validator

- Upload a payment receipt image.
- OCR extracts available payment details.
- Matching rules compare extracted details against unpaid orders.
- Potential duplicates, mismatched amounts and incomplete receipts are flagged.

AI only where it adds value: AI assists with extraction. Rules handle matching. The owner reviews the result before confirming payment.

HOW IT WORKS

One complete journey, end to end

STEP 1

Trigger

A WhatsApp order is pasted. Staff upload payment slips.

STEP 2

Connected data

Menu, unpaid orders and the bank statement.

STEP 3

Decision

AI reads. Rules match slip, order and bank, then sort by colour.

STEP 4

Human approval

Only the owner approves, rejects or sends a reminder.

STEP 5

Outcome

Order status updates and every action is saved in the audit log.

HOW IT WORKS

From WhatsApp message to verified payment

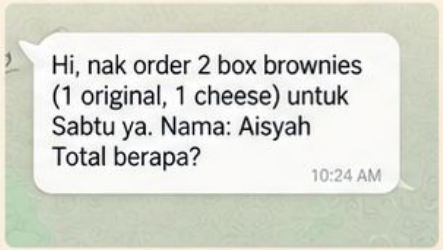
One complete journey, end to end.

1 Chat to bill

Turn a WhatsApp message into a ready-to-pay bill.

1 Paste WhatsApp message

Staff pastes the customer's order message in BM, English or Manglish.



2 AI extracts order details

AI identifies customer name, items, quantities, flavours and date. Missing information is flagged.

Extracted order

Customer	Aisyah binti Rahman
Items	Brownies (Original) × 1 Brownies (Cheese) × 1
Date	25 Oct 2026

✓ All details captured

3 Automatically generate order list & bill

System matches items with menu prices, calculates total and creates an order ID and bill.

Order ORD-261025-0043 — Pending payment

Item	Qty	Price (RM)	Total (RM)
Brownies (Original)	1	32.00	32.00
Brownies (Cheese)	1	32.00	32.00
Total			**RM 65.00**

2 Payment validation

Read the receipt and match it with the order and bank record.

4 Upload payment receipt

Staff uploads the customer's bank or e-wallet payment slip.



5 AI reads the receipt

Runpod OCR extracts the amount, reference number, sender and date.

Extracted from receipt

Amount	RM 65.00
Reference	DB2610009
Sender	AISYAH BINTI RAHMAN
Date	20 Oct 2026

✓ Readable (confidence 96%)

6 Three-way payment matching

Rules compare the receipt, generated order and imported bank statement.

Receipt (from Step 5) — Order (from Step 3) — Bank statement (imported)

Result

- ✓ **Green · Verified match** — All key details match
- ! **Yellow · Needs review** — Partial match or uncertain data
- ✕ **Red · Risk detected** — Mismatch, duplicate or suspicious

3 Owner decision

Only the owner approves, and every action is saved in the audit log.

7 Owner reviews the result

Owner checks the extracted details and matching result. Approve, reject or send a short-payment reminder.

Order ID	ORD-261025-0043
Customer	Aisyah binti Rahman
Amount	RM 65.00
Reference	DB2610009
Result	Yellow · Needs review

✓ Approve | ✕ Reject | ↗ Send reminder

8 Update order status & audit log

System updates the order status and records the owner's action in the audit log.

Order updated
- ✓ Payment status updated
- ✓ Order marked as Paid / Unpaid
- ✓ Customer notified (if needed)

+

Audit log recorded
- ✓ Who made the decision
- ✓ What action was taken
- ✓ Timestamp and notes

THE THREE-WAY MATCH

No real money, no green light

Slip: what the customer claims **vs** **Order**: what they should pay **vs** **Bank**: what really arrived

#	Check, in order	If it fails
1	Is it a readable payment slip (confidence 80% or more)?	Manual: human eye
2	Is there a bank credit with this reference?	Red: no money arrived
3	Is that credit still unused by any other slip?	Red: duplicate or reused ref
4	Does the slip amount equal the bank amount?	Red: edited slip
5	Does the bank amount equal the amount due?	Yellow: short or over pay

All five pass: **Green**. Combo-Math can add up several orders from one customer.

TRAFFIC-LIGHT KANBAN

The owner sees three clear columns

Green: Verified

Slip, order and bank all match. Bulk approve in one click.

Demo: Aisyah paid RM65 for ORD-1001

Yellow: Check

Short or over pay, or a cash deposit with no name.

Demo: Nurul paid RM130 of RM150. A WhatsApp reminder is ready.

Red: Fraud risk

No bank credit, edited amount, or a reference already used.

Demo: Chong’s slip reuses Farah’s bank ref

Live demo

ORDER → SLIP → CHECK → APPROVE → LOG



PEOPLE STAY IN CONTROL

Staff upload. Only the owner approves.

Role-based access (simulated)

Action	Staff	Owner
Upload payment slips	Yes	Yes
Upload bank statement	No	Yes
Approve or reject	No	Yes
Send short-pay reminder	No	Yes
View audit log	No	Yes

Audit trail (example entries)

10:15 Staff uploaded slip S01

10:15 AI read slip: RM65, ref DB2610001

10:15 Matched ORD-1001 and bank credit: Green

10:20 Owner approved: ORD-1001 Paid

16:44 Slip flagged Red: ref already used

TECH STACK

Built with the Phase 1 toolkit

Layer	Technology	Role
Frontend	HTML, CSS and JavaScript	Dashboard, forms and review queue
Backend	Python and FastAPI	API endpoints and application logic
Chat-to-Bill AI	Python parsing and validation rules	Reads WhatsApp messages into clean orders
OCR / Vision AI	RunPod Serverless · [model]	Reads slips, flags editing, gives a confidence score
Matching logic	Team-written rules	Three-way match, Combo-Math, short pay, used-credit check
Database	SQLite	Menu, orders, slips and audit log
Bank records	Synthetic bank statement (CSV)	Proof of what money really arrived
Messaging	WhatsApp click-to-chat link	Pre-filled short-pay reminder, sent by the owner
Access control	Simulated login	Staff and Owner roles
Code	GitHub	https://github.com/xian3512/SDC2026---Catalyst/, public for judges

SDC sponsor platforms: Replit builds and hosts the app · RunPod Serverless hosts the OCR / Vision AI

IMPACT AND KPIs

How we measure it

Synthetic test set

- 15 orders from 18 WhatsApp messages
- 15 payment slips: bank, e-wallet, cash deposit, and a cat photo
- 1 bank statement with 11 credits
- 4 seeded fake slips

KPI	Phase 2 target	Result
Time from slip upload to verified match	85% faster	88% faster
Owner time to verify one slip	50% less or better	60% less
Reused refs and short pays detected	100%	100% (4/4 test cases)
Yellow and Red actions approved and logged	100%	100% (all test actions)

Baseline: 14,881 e-commerce fraud cases, RM123.7M lost (PDRM, Jan–Nov 2025) · ~1% manual data-entry error (Barchard & Pace, 2011)

SDG 8 AND SDG 9

Helping small sellers grow safely

Target 8.2

Higher productivity through technological upgrading and innovation

Replaces hand-checking of orders and slips with AI and simple rules

Target 8.3

Encourage the formalization and growth of micro, small and medium enterprises

Protects cash flow and creates clean digital payment records

Target 9.3

Small enterprises' access to financial services and markets

Makes digital payments safer to accept through the Orbyte hub

Source: UN SDG targets (metadata.un.org/sdg/8) · UN SDG metadata for target 9.3 (World Bank)

ORBYTE FIT AND REUSE

One module, ready for any SME on Orbyte

Fits the hub

- Plug in any menu or product list
- Orders, payments and approvals in one place
- Works for bakeries, clothing shops, any seller paid by transfer
- Easy to learn: paste, upload, approve

Roadmap after the prelim

- Live bank feed (future assumption)
- Daily capacity check, no overbooking
- Auto-reply asking for missing order info
- Stronger visual tamper detection

RISKS AND PRIVACY

We planned for what could go wrong

Risk	Our plan
Hardest risk: AI misreads a blurry or cropped slip	The bank check confirms every payment. Below 80% confidence, the slip goes to a human.
Vision AI is down	All slips go to the human queue. The journey still completes.
Chat-to-Bill fails in the demo	A “Load sample orders” button keeps the payment flow running.

Privacy and PDPA

- Demo uses only team-made synthetic data
- Real use follows PDPA’s seven principles, including retention
- Amended PDPA (from 1 June 2025): data protection officer and breach notification
- Roles limit who sees payment data; the audit log records every action

Source: ASEAN Briefing (2025), “Malaysia Tightens Data Protection from June 2025”

Less checking.
More baking.

From WhatsApp chat to verified payment, in one click.

GitHub: https://github.com/xian3512/SDC2026---Catalyst/

Thank you!